Bibliografía. Rendimiento de cebada de temporal con régimen de registro y rezago espacial

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Referencias del informe y del anexo técnico, con la numeración que usan ambos documentos. Están en el orden en que se citan por primera vez, primero en el informe y después en el anexo.

Referencias

[1] FIRA, *Reto AgroCebada 2026. Bases de participación y descripción del conjunto de datos*, 2026.

[2] Secretaría de Agricultura y Desarrollo Rural, "Del campo a la copa: el recorrido de la cebada," *gob.mx*, 11 de noviembre de 2025. [En línea]. Disponible: https://www.gob.mx/agricultura/articulos/del-campo-a-la-copa-el-recorrido-de-la-cebada

[3] FIRA, *Sesiones de preguntas y respuestas del Reto AgroCebada 2026 y correos de seguimiento a los equipos participantes*, septiembre y octubre de 2026 (último correo del 14 de octubre).

[4] M. Drusch *et al.*, "Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012.

[5] A. E. Frazier y B. L. Hemingway, "A Technical Review of Planet Smallsat Data: Practical Considerations for Processing and Using PlanetScope Imagery," *Remote Sensing*, vol. 13, no. 19, p. 3930, 2021.

[6] C. Funk *et al.*, "The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes," *Scientific Data*, vol. 2, art. 150066, 2015.

[7] Climate Hazards Center, "CHIRTS-ERA5," conjunto de datos, Universidad de California en Santa Bárbara, 2025. [En línea]. Disponible: https://doi.org/10.15780/G2F08J

[8] D. B. Lobell *et al.*, "Eyes in the Sky, Boots on the Ground: Assessing Satellite- and Ground-Based Approaches to Crop Yield Measurement and Analysis," *American Journal of Agricultural Economics*, vol. 102, no. 1, pp. 202–219, 2020.

[9] T. van Klompenburg, A. Kassahun y C. Catal, "Crop yield prediction using machine learning: A systematic literature review," *Computers and Electronics in Agriculture*, vol. 177, p. 105709, 2020.

[10] M. Weiss, F. Jacob y G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sensing of Environment*, vol. 236, p. 111402, 2020.

[11] J. A. Pomares Valdés, V. J. López Martínez y L. E. Ramiro Luna, "¿Qué separa a los dos regímenes de rendimiento? Descarte de causas ambientales y contraste con las cifras oficiales del SIAP," Benemérita Universidad Autónoma de Puebla, Puebla, México, Rep. técnico, oct. 2026. [En línea]. Disponible: https://github.com/alan-pv/AgroCebada_Hackaton/blob/main/docs/reportes/REGIMENES.pdf

[12] J. A. Pomares Valdés, V. J. López Martínez y L. E. Ramiro Luna, "El Reto AgroCebada 2026 en contexto. Lo que sabemos, lo que suponemos y lo que falta comprobar," Benemérita Universidad Autónoma de Puebla, Puebla, México, Rep. técnico, oct. 2026. [En línea]. Disponible: https://github.com/alan-pv/AgroCebada_Hackaton/blob/main/docs/reportes/CONTEXTO.pdf

[13] J. A. Pomares Valdés, V. J. López Martínez y L. E. Ramiro Luna, "Rendimiento de cebada de temporal con régimen de etiquetas y rezago espacial. Procedimiento, arquitectura y evaluación del modelo," Benemérita Universidad Autónoma de Puebla, Puebla, México, Rep. técnico, oct. 2026. [En línea]. Disponible: https://github.com/alan-pv/AgroCebada_Hackaton/blob/main/docs/reportes/MODELO.pdf

[14] J. A. Pomares Valdés, V. J. López Martínez y L. E. Ramiro Luna, "¿Dónde rinde más la cebada de temporal? Un modelo municipal con datos públicos para Hidalgo, México, Puebla y Tlaxcala," Benemérita Universidad Autónoma de Puebla, Puebla, México, Rep. técnico, oct. 2026. [En línea]. Disponible: https://github.com/alan-pv/AgroCebada_Hackaton/blob/main/docs/reportes/MUNICIPAL.pdf

[15] J. A. Pomares Valdés, V. J. López Martínez y L. E. Ramiro Luna, "AgroCebada. Código, datos públicos y reportes del Reto AgroCebada FIRA 2026," GitHub, 2026. [En línea]. Disponible: https://github.com/alan-pv/AgroCebada_Hackaton

[16] FIRA, *Ecuaciones de índices satelitales. Reto AgroCebada FIRA 2026*, 2026.

[17] J. C. Zadoks, T. T. Chang y C. F. Konzak, "A decimal code for the growth stages of cereals," *Weed Research*, vol. 14, no. 6, pp. 415–421, 1974.

[18] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[19] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nature Communications*, vol. 11, no. 1, art. 4540, 2020.

[20] M. Ojala y G. C. Garriga, "Permutation tests for studying classifier performance," *Journal of Machine Learning Research*, vol. 11, pp. 1833–1863, 2010.

[21] W. R. Tobler, "A computer movie simulating urban growth in the Detroit region," *Economic Geography*, vol. 46, pp. 234–240, 1970.

[22] C. E. Rasmussen y C. K. I. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA, EE. UU.: MIT Press, 2006.

[23] G. C. Cawley y N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010.

[24] S. Varma y R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, art. 91, 2006.

[25] V. Vovk, A. Gammerman y G. Shafer, *Algorithmic Learning in a Random World*. Nueva York, NY, EE. UU.: Springer, 2005.

[26] J. Lei *et al.*, "Distribution-Free Predictive Inference for Regression," *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1094–1111, 2018.

[27] S. Kaufman, S. Rosset, C. Perlich y O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, art. 15, 2012.

[28] Servicio de Información Agroalimentaria y Pesquera, "Cierre de la producción agrícola por municipio, cebada grano, temporal, primavera-verano, 2003–2025." [En línea]. Disponible: https://nube.agricultura.gob.mx/cierre_agricola/. Consultado el 9 de octubre de 2026.

[29] V. S. Passarella, R. Savin y G. A. Slafer, "Grain weight and malting quality in barley as affected by brief periods of increased spike temperature under field conditions," *Australian Journal of Agricultural Research*, vol. 53, no. 11, pp. 1219–1227, 2002.

[30] M. A. Oliver y R. Webster, "A tutorial guide to geostatistics: Computing and modelling variograms and kriging," *CATENA*, vol. 113, pp. 56–69, 2014.

[31] Fideicomisos Instituidos en Relación con la Agricultura, "Agrocostos." [En línea]. Disponible: https://www.fira.gob.mx/Nd/Agrocostos.jsp. Consultado el 15 de octubre de 2026.

[32] Y. Gu *et al.*, "A five-year analysis of MODIS NDVI and NDWI for grassland drought assessment over the central Great Plains of the United

States," *Geophysical Research Letters*, vol. 34, no. 6, art. L06407, 2007, doi: 10.1029/2006GL029127.
[33] N. Jensen y C. Barrett, "Agricultural Index Insurance for Development," *Applied Economic Perspectives and Policy*, vol. 39, no. 2, pp. 199–219, 2017.
[34] T. Chai y R. R. Draxler, "Root mean square error (RMSE) or mean absolute error (MAE)? Arguments against avoiding RMSE in the literature," *Geoscientific Model Development*, vol. 7, no. 3, pp. 1247–1250, 2014.
[35] T. Hastie, R. Tibshirani y J. Friedman, *The Elements of Statistical Learning. Data Mining, Inference, and Prediction*, 2a ed. Nueva York, NY, EE. UU.: Springer, 2009.
[36] D. Shepard, "A two-dimensional interpolation function for irregularly-spaced data," en *Proc. 23rd ACM National Conference*, 1968, pp. 517–524.
[37] R. L. Wasserstein y N. A. Lazar, "The ASA statement on p-values. Context, process, and purpose," *The American Statistician*, vol. 70, no. 2, pp. 129–133, 2016.
[38] Y. Benjamini y Y. Hochberg, "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 57, no. 1, pp. 289–300, 1995.
[39] C. Nadeau e Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
[40] B. Efron y R. J. Tibshirani, *An Introduction to the Bootstrap*. Nueva York, NY, EE. UU.: Chapman & Hall, 1993.
[41] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
[42] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sensing of Environment*, vol. 8, no. 2, pp. 127–150, 1979.
[43] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society, Series B*, vol. 58, no. 1, pp. 267–288, 1996.
[44] A. E. Hoerl y R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
[45] H. Meyer y E. Pebesma, "Predicting into unknown space? Estimating the area of applicability of spatial prediction models," *Methods in Ecology and Evolution*, vol. 12, no. 9, pp. 1620–1633, 2021.
[46] Instituto Nacional de Estadística y Geografía, "Índice Nacional de Precios al Consumidor (INPC)." [En línea]. Disponible: https://www.inegi.org.mx/temas/inpc/
[47] Anthropic, "Claude Code." [En línea]. Disponible: https://www.anthropic.com/claude-code. Consultado el 15 de octubre de 2026.
[48] J. A. Pomares Valdés, V. J. López Martínez y L. E. Ramiro Luna, "Prompts principales del proyecto AgroCebada," registro de uso de inteligencia artificial, GitHub, oct. 2026. [En línea]. Disponible: https://github.com/alan-pv/AgroCebada_Hackaton/blob/main/docs/PROMPTS.md